



NORTHERN BEACHES SECONDARY COLLEGE

MANLY SELECTIVE CAMPUS

Year 12

Trial Examination

2022

Mathematics Extension II

General Instructions

- Reading time – 10 minutes
- Working time – 3 hours
- Write using black pen
- Write your name on the front of every booklet.
- In Questions 11 to 16 show relevant mathematical reasoning and/or calculations.
- NESA approved calculators and templates may be used.
- Weighting: 30%

Section I Multiple Choice

- 10 marks
- Attempt all questions.
- Answer Sheet provided
- Allow about 15 minutes for this section

Section II Free Response

- 90 marks
- Start a separate booklet for each question.
- Each question is of equal value.
- All necessary working should be shown in every question.
- Allow about 2 hours and 45 minutes for this section.

Section I

10 marks

Attempt Questions 1 – 10

Allow about 15 minutes for this section

Use the multiple-choice answer sheet for Questions 1 – 10.

Q1. Given that $1 - i$ is a solution of $z^3 - 4z^2 + 6z - 4 = 0$, what are the other solutions?

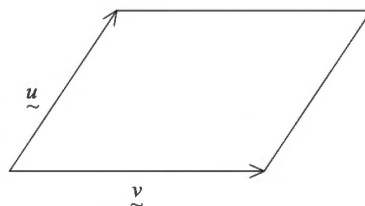
- A. $1 + i$ and -6
- B. $1 + i$ and 2
- C. $-1 - i$ and $4 + 2i$
- D. $1 + i$ and -4

Q2. Let $\underline{u} = 2\underline{i} - a\underline{j} - \underline{k}$ and $\underline{v} = 3\underline{i} + 2\underline{j} - b\underline{k}$.

If these vectors are perpendicular to each other, what are the possible values for a and b ?

- A. $a = 2$ and $b = 2$
- B. $a = -2$ and $b = 10$
- C. $a = 0$ and $b = 0$
- D. $a = -1$ and $b = -8$

Q3. The diagram below shows a rhombus.

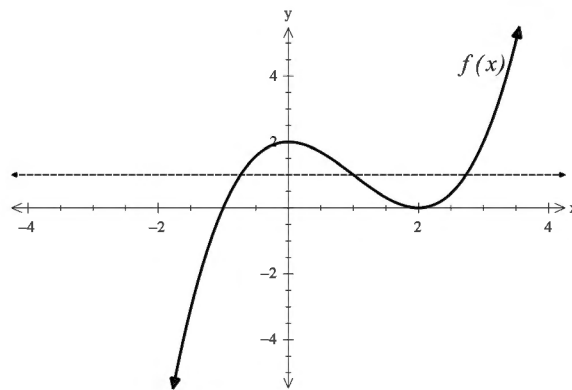


Adjacent sides are represented by two vectors, $\underline{\tilde{u}}$ and $\underline{\tilde{v}}$.

It follows that:

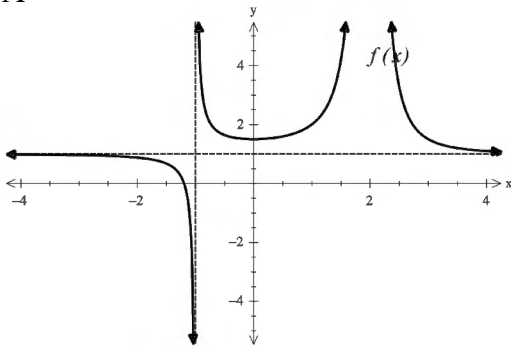
- A. $\underline{\tilde{u}} \cdot \underline{\tilde{v}} = 0$
- B. $\underline{\tilde{u}} = \underline{\tilde{v}}$
- C. $(\underline{\tilde{u}} + \underline{\tilde{v}}) \cdot (\underline{\tilde{u}} - \underline{\tilde{v}}) = 0$
- D. $|\underline{\tilde{u}} + \underline{\tilde{v}}| = |\underline{\tilde{u}} - \underline{\tilde{v}}|$

Q4. The graph of the function $y = f(x)$ is shown below.

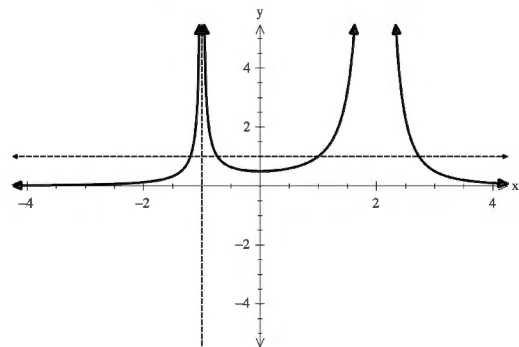


Which graph below shows the most accurate representation of the graph $y = \frac{1}{f(x)}$?

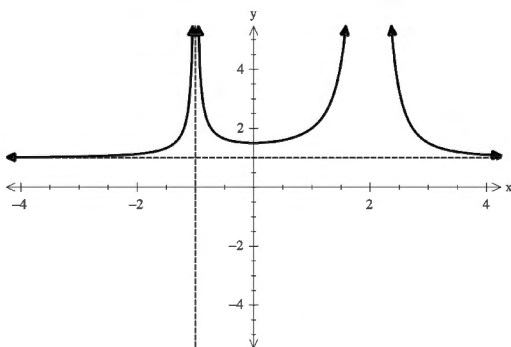
A



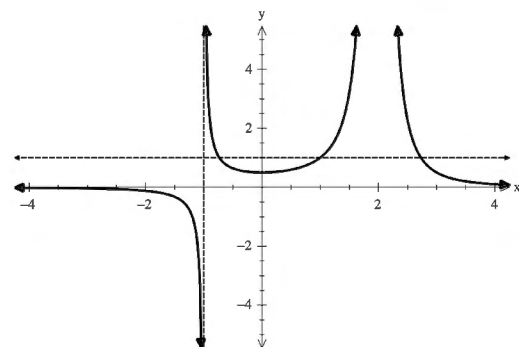
B



C



D



Q5. Consider the statement:

If the soup contains lemon or vinegar, the soup does not contain chilli.

Which of the following is logically equivalent to the statement?

- A. If the soup contains chilli, the soup does not contain lemon nor vinegar.
- B. If the soup contains chilli, the soup does not contain lemon or the soup does not contain vinegar.
- C. If the soup does not contain lemon or the soup does not contain vinegar, the soup contains chilli.
- D. If the soup does not contain lemon nor vinegar, the soup contains chilli.

Q6. The vector equation of the line that passes through the points A (1, 0, 2) and B (3, 9, 6) is given by which of the following?

- A. $\underline{r} = \underline{i} + 2\underline{k} + \lambda(2\underline{i} + 9\underline{j} + 4\underline{k})$
- B. $\underline{r} = \underline{i} + 2\underline{k} + \lambda(2\underline{i} + 9\underline{j} + 8\underline{k})$
- C. $\underline{r} = \underline{i} + 2\underline{k} + \lambda(4\underline{i} + 9\underline{j} + 4\underline{k})$
- D. $\underline{r} = \underline{i} + 2\underline{k} + \lambda(2\underline{i} + 4\underline{k})$

Q7. Which expression is equal to $\int \tan^{-1} x \, dx$?

- A. $x \tan^{-1} x - \int \frac{x}{1+x^2} dx$
- B. $x^2 \tan^{-1} x - \int \frac{x^2}{1+x^2} dx$
- C. $\frac{1}{2}(\tan^{-1} x)^2$
- D. $\frac{x}{2}(\tan^{-1} x)^2 - \int \frac{x}{\tan x} dx$

Q8. Which of the following statements is FALSE?

- A. $\forall x \in \mathbb{R}, \exists y \in \mathbb{R}$ such that $x^4 = 2y^3$
- B. $\forall x \in \mathbb{R}, \exists y \in \mathbb{R}$ such that $x = y \Leftrightarrow \sin x = \sin y$
- C. $\exists x \in \mathbb{R}, \forall y \in \mathbb{R}$ such that $\sqrt{x} < e^y$
- D. $\exists x \in \mathbb{R}, \forall y \in \mathbb{R}$ such that $|y| > x$

Q9. $\int_{-1}^2 2x\sqrt{x+2} \, dx$.

- A. $\frac{46}{15}$
- B. $\frac{92}{15}$
- C. $\frac{113}{2}$
- D. $\frac{28+32\sqrt{2}}{15}$

Q10. If $\int_a^b \frac{1}{\sin x + \cos x} \, dx = 1$, where $a, b \in \mathbb{R}$,

what is the value of $\int_{2a}^{2b} \frac{1}{\sin \frac{x}{2} + \cos \frac{x}{2}} \, dx$

- A. $\frac{1}{4}$
- B. $\frac{1}{2}$
- C. 1
- D. 2

End of Multiple Choice

Section II

90 marks

Attempt Questions 11 – 16

Allow about 2 hours and 45 minutes for this section

Answer each question in the appropriate writing booklet. Extra writing booklets are available.

In Questions 11 – 16, your responses should include relevant mathematical reasoning and/or calculations.

Question 11

15 marks

a. Let $z = 4cis\frac{5\pi}{6}$ and $w = \sqrt{2}cis\frac{\pi}{4}$.

(i) Find z^7 in modulus-argument form.

2

(ii) Express $\frac{z}{w}$ in both modulus-argument form and Cartesian form.

3

(iii) Hence or otherwise, find the exact value of $\cos\frac{7\pi}{12}$.

2

b. A point P moves on the x -axis so that its x -coordinate, in metres, at time t seconds is given by

$$x = 8[1 - \sin(2t)] + 3\cos(2t)$$

(i) Show that the motion of P is simple harmonic.

2

(ii) State centre and period of the motion of P .

2

c. Sketch the region on the Argand diagram defined by the following

$$\{z: 1 < |z| \leq 3\} \cap \{z: \operatorname{Im}(z) > \operatorname{Re}(z)\}$$

4

End of Question 11

- a. (i) Find the values of a, b and c such that 3

$$\frac{5x}{(x-1)(x^2+2x+2)} \equiv \frac{a}{x-1} + \frac{bx+c}{x^2+2x+2}$$

- (ii) Hence, or otherwise, find 3

$$\int \frac{5x}{(x-1)(x^2+2x+2)} dx.$$

- b. Let $\omega = e^{i(\pi/n)}$, where ω is a complex number ie. ω is not $\in \mathbb{R}$

- (i) Show that ω is a solution of $z^{2n} = 1$. 1

- (ii) Given $a^n - b^n = (a-b)(a^{n-1} + a^{n-2}b + \dots + b^{n-1})$,
deduce that $1 + \omega + \dots + \omega^{2n-1} = 0$. 1

- (iii) Show that $1 + \omega + \dots + \omega^{n-1} = 1 + i \cot\left(\frac{\pi}{2n}\right)$. 3

- c. Prove the following statement by contradiction. 4

*If a, b and c are odd integers and $x \in \mathbb{Z}$, then the equation
 $ax^2 + bx + c = 0$ has no integer solution.*

End of Question 12

Question 13**15 marks**

- a. (i) Express $1 + i \cot \theta$ in modulus-argument form, where $0 < \theta < \frac{\pi}{2}$ 2
- (ii) Hence, or otherwise, simplify $(1 + i \cot \theta)^{-3}$ 1
- b. Find the exact value of $\int_0^{\frac{\pi}{2}} \frac{1}{2 - \sin x} dx$. 4
- c. A sequence is defined recursively as $u_1 = 1$, $u_{n+1} = 3u_n + 2n + 1$, for all positive integers n .
] Use Mathematical Induction to prove that $u_n = 3^n - n - 1$ for all positive integers n . 3
- d. A particle is initially at rest on the number line at a position of $x = 1$. The particle moves continuously along the number line according to the acceleration equation
- $$\ddot{x} = \frac{4}{(x-2)^2} + \frac{8}{x^3}.$$
- At time t , the velocity of the particle is v .
- (i) Show that $v^2 = -\frac{8}{x-2} - \frac{8}{x^2}$ 2
- (ii) Hence, or otherwise, determine the possible range of particle's displacement as it moves along the number line. 3

Question 13 completed.

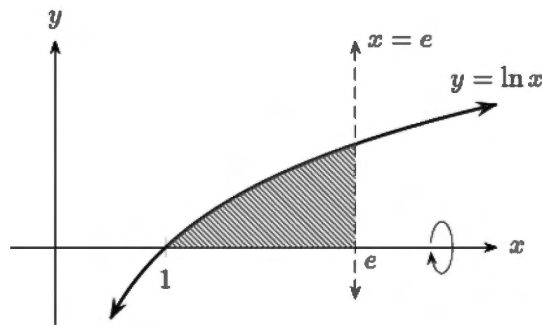
- a. Use an appropriate substitution to evaluate $\int_0^4 \frac{\sqrt{x}}{(x\sqrt{x}+1)^2} dx$ 3
- b. Consider the statement about two positive integers m and n .
 “If $m^2 + n^2$ is even, then either both m and n are even or both m and n are odd.”
- (i) Write down the contrapositive of the statement. 1
- (ii) Prove the original statement by contraposition. 2
- c. (i) If $0 < a < 4$, prove that $\frac{1}{a} + \frac{1}{4-a} \geq 1$. 2
- (ii) If $0 < a < 4, 0 < b < 4$ and $0 < c < 4$, prove that at least one of the numbers
- $$\frac{1}{a} + \frac{1}{4-b}, \frac{1}{b} + \frac{1}{4-c}, \frac{1}{c} + \frac{1}{4-a}$$
- is greater than or equal to one. 3
- d. Let $I_n = \int_0^1 \frac{x^{2n}}{\sqrt{1+x^2}} dx$, where $n = 1, 2, 3, \dots$ 4
- Deduce that $2nI_n + (2n-1)I_{n-1} = \sqrt{2}$, where $n = 2, 3, 4, \dots$

Question 14 completed.

Question 15

15 marks

- a. The region enclosed by the curve $y = \ln x$, the line $x = e$ and the x axis is shown in the diagram. **3**



Find the volume of the solid formed by rotating the shaded region about the x axis.

- b. Sphere S has a vector equation

$$\left| \vec{r} - (3\vec{i} + \vec{j} + 4\vec{k}) \right| = \sqrt{35}$$

- (i) Write the Cartesian equation of this sphere. **1**

- (ii) The line l has the equation $\vec{r} = \begin{pmatrix} 2 \\ 0 \\ 3 \end{pmatrix} + \lambda \begin{pmatrix} 1 \\ 2 \\ -1 \end{pmatrix}$, $\lambda \in \mathbb{R}$. **3**

Determine whether this line is a tangent to the sphere S or not. Give full reasons for your answer.

- c. Show there exists complex numbers z and w , such that **3**

$$\frac{1}{z} + \frac{1}{w} = \frac{1}{z+w} \text{ where } z \neq 0 \text{ and } w \neq 0.$$

- d. (i) Show $a^2 + b^2 + c^2 \geq ab + ac + bc$ **1**

- (ii) Hence, show $(a + b + c)^2 \geq 3(ab + ac + bc)$ **1**

- (iii) Hence, prove $a^2 + b^2 + c^2 \geq 3 \times \sqrt[3]{a^2 b^2 c^2}$ **3**

$$(\text{Given } a^3 + b^3 = (a + b)(a^2 - ab + b^2))$$

Question 15 completed

Question 16

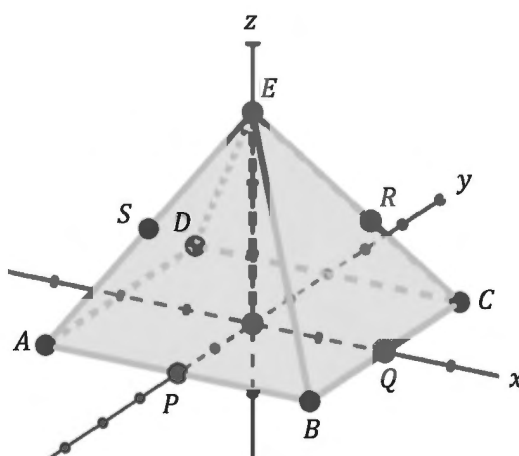
15 marks

- a. (i) Solve $z^3 = -1$, giving your answers in modulus-argument form. 2

- (ii) Using part (i), or otherwise, find the solutions to the equation 3

$$(z^2 + 2)^3 = -1$$

- b. ABCDE is a pyramid with base measuring 8 cm by 8 cm.
The height of the pyramid is $4e$, where e is a real positive number.
The origin O is the centre of the base.



The points P, Q, R and S are the midpoints of sides AB, BC, CE and EA respectively.

The position vectors of A, B, C and D relative to O , respectively, are

$$-4\underset{\sim}{i} - 4\underset{\sim}{j}, 4\underset{\sim}{i} - 4\underset{\sim}{j}, 4\underset{\sim}{i} + 4\underset{\sim}{j} \text{ and } -4\underset{\sim}{i} + 4\underset{\sim}{j}.$$

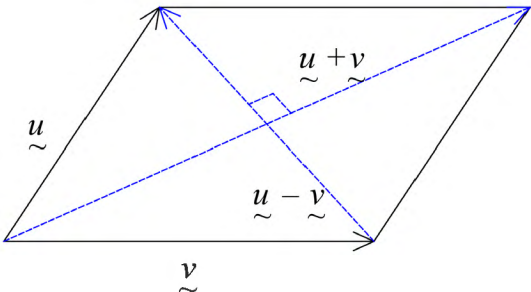
- (i) Show, in terms of e , that $\overrightarrow{PR} = 2\underset{\sim}{i} + 6\underset{\sim}{j} + 2e\underset{\sim}{k}$ and $\overrightarrow{QS} = -6\underset{\sim}{i} - 2\underset{\sim}{j} + 2e\underset{\sim}{k}$. 3

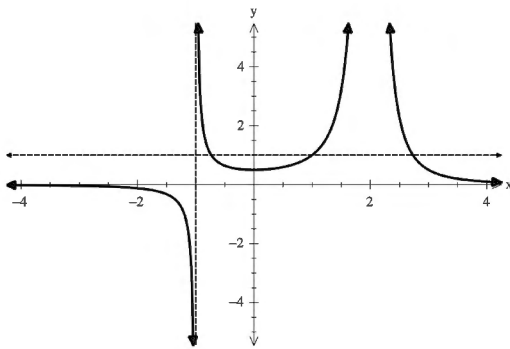
- (ii) Hence or otherwise, write the vector equations of line l , passing through points P and R , and line s , passing through points Q and S . 2

- (iii) Find the position vector $\overrightarrow{OM}$ in terms of e , where M is the point of intersection of lines l and s . 2

- (iv) Given that $\overrightarrow{OM}$ is perpendicular to $\overrightarrow{EB}$, find the value of e and hence find the acute angle between $\overrightarrow{PR}$ and $\overrightarrow{QS}$. Give your answer to the nearest degree. 3

End of paper

Q1	$z^3 - 4z^2 + 6z - 4 = 0$ roots are $(1 - i)$ and $(1 + i)$ factor $z^2 - 2\operatorname{Re}(\alpha)z + \alpha ^2$ $\therefore z^2 - 2z + 2$ $z^3 - 4z^2 + 6z - 4 = (z^2 - 2z + 2)(az + b)$ $\therefore \begin{aligned} a &= 1 \\ b &= -2 \end{aligned}$ Other roots are $1 + i$ and 2	B
Q2	$\begin{aligned} u &= 2i - aj - k \\ v &= 3i + 2j - bk \end{aligned}$ $u \cdot v = 0$ $u \cdot v = 2 \times 3 - 2a + b = 0$ $b - 2a = -6$ $-8 - 2(-1) = -6$ $a = -1 \quad b = -8$	D
Q3	 Diagonals of rhombus are perpendicular therefore $(\underline{\underline{u}} + \underline{\underline{v}}) \cdot (\underline{\underline{u}} - \underline{\underline{v}}) = 0$	C

Q4		D
Q5	If the soup contains chilli, the soup does not contain lemon nor vinegar.	A
Q6	$A = \underset{\sim}{i} + 2\underset{\sim}{k}$ $B = 3\underset{\sim}{i} + 9\underset{\sim}{j} + 6\underset{\sim}{k}$ $\overrightarrow{AB} = 2\underset{\sim}{i} + 9\underset{\sim}{j} + 4\underset{\sim}{k}$ $\underset{\sim}{r} = \underset{\sim}{i} + 2\underset{\sim}{k} + \lambda \left(2\underset{\sim}{i} + 9\underset{\sim}{j} + 4\underset{\sim}{k} \right)$	A
Q7	$u = \tan^{-1} x \qquad v' = 1$ $u' = \frac{1}{1+x^2} \qquad v = x$ $I = uv - \int vu' dx$ $= x \tan^{-1} x - \int \frac{x}{1+x^2} dx$	A
Q8	$\forall x \in \mathbb{R}, \exists y \in \mathbb{R}$ such that $x = y \Leftrightarrow \sin x = \sin y$	B

Q9	$\int_{-1}^2 2x\sqrt{x+2} \, dx$ $u = x + 2$ $x = u - 2 \Rightarrow dx = du$ $x = -1 \Rightarrow u = 1$ $x = 2 \Rightarrow u = 4$ $\int_1^4 2(u-2)\sqrt{u} \, du$ $= 2 \int_1^4 u^{\frac{3}{2}} - 2u^{\frac{1}{2}} \, du$ $= 2 \left[\frac{2}{5} u^{\frac{5}{2}} - \frac{4}{3} u^{\frac{3}{2}} \right]_1^4$ $= 2 \left\{ \left(\frac{2}{5} \times 2^5 - \frac{4}{3} \times 2^3 \right) - \left(\frac{2}{5} \times 1 - \frac{4}{3} \times 1 \right) \right\}$ $= 2 \left(\frac{64}{5} - \frac{32}{3} - \frac{2}{5} + \frac{4}{3} \right) = \frac{92}{15}$	A
Q10	$\int_a^b \frac{1}{\sin x + \cos x} \, dx = 1$ $\text{let } x = \frac{u}{2} \Rightarrow 2x = u$ $2dx = du$ $x = a \Rightarrow u = 2a$ $x = b \Rightarrow u = 2b$ $\therefore \int_{2a}^{2b} \frac{2}{\sin \frac{u}{2} + \cos \frac{u}{2}} \, du = 1$ $\int_{2a}^{2b} \frac{1}{\sin \frac{x}{2} + \cos \frac{x}{2}} \, dx = \frac{1}{2}$	B

Question 11.

a-i	$z = R\text{cis}\theta$ $z^7 = R^7\text{cis}(7\theta)$ $\therefore z^7 = 4^7\text{cis}\frac{35\pi}{6}$ $= 16384\text{cis}\left(-\frac{\pi}{6}\right)$	2 marks – correct solution 1 mark – either - correct argument - correct modulus
a-ii	ModArg form $\frac{z_1}{z_2} = \frac{R_1\text{cis}\theta}{R_2\text{cis}\beta} = \frac{R_1}{R_2}\text{cis}(\theta - \beta)$ $= \frac{4}{\sqrt{2}}\text{cis}\left(\frac{5\pi}{6} - \frac{\pi}{4}\right)$ $= 2\sqrt{2}\text{cis}\frac{7\pi}{12}$ Cartesian form: $\frac{z}{w} = \frac{-2\sqrt{3} + 2i}{1 + i} \times \frac{1 - i}{1 - i} = (1 - \sqrt{3}) + i(1 + \sqrt{3})$ $\frac{z}{w} = \frac{-2\sqrt{3} + 2}{2} + \frac{2 + 2\sqrt{3}}{2}i \quad \text{was also accepted}$ Decimal Form was <i>NOT</i> accepted	3 marks Provides correct solution 2 marks Expresses fraction in both modulus-argument or Cartesian form (but with the denominator not realised), or equivalent merit 1 mark - uses $\frac{z_1}{z_2}$ to produce correct modulus or argument - uses $\frac{z}{w}$ to produce correct fraction
a-iii	Using part (ii) and equating the real parts, $2\sqrt{2}\cos\frac{7\pi}{12} = 1 - \sqrt{3}$ $\therefore \cos\frac{7\pi}{12} = \frac{1 - \sqrt{3}}{2\sqrt{2}}$	2 marks – correct solution 1 mark - Attempts to equate real or imaginary parts of the two expression in part (i), or equivalent merit

c-i	Note: $x = 8[1 - \sin(2t)] + 3\cos(2t)$ $\therefore x - 8 = -8\sin(2t) + 3\cos(2t)$ $\dot{x} = -16\cos(2t) - 6\sin(2t)$ $\ddot{x} = 32\sin(2t) - 12\cos(2t)$ $= -4(x - 8) \text{ which is in the form } -n^2(x - c)$ Hence motion is SHM	2 marks – correct solution 1 mark – correct approach with one arithmetic error.
c-ii	Centre $x = 8$, period $= \frac{2\pi}{2} = \pi$	2 marks – both answers 1 mark – one answer only
d	Features to include 1. Correct circles and shading. 2. Correct boundaries ie. full or dotted line 3. Correct angle or indicated line is $y = x$ or equivalent, 4. Open circles used correctly.	4 marks – one mark for each correct feature

Question 12

a-i	$5x \equiv a(x^2 + 2x + 2) + (x - 1)(bx + c)$ Let $\begin{array}{lll} x = 1: & 5 = 5a & \Rightarrow a = 1 \\ x = 0: & 0 = 2a - c & \Rightarrow c = 2 \\ x = -1: & -5 = a - 2(-b + c) & \Rightarrow b = -1 \end{array}$	3marks – correct solution 2 marks – correct approach with one arithmetic error. 1 mark – correct initial equation equating numerators
a-ii	$I = \int \frac{5x}{(x-1)(x^2+2x+2)} dx = \int \left(\frac{1}{x-1} + \frac{-x+2}{x^2+2x+2} \right) dx$ $I = \int \left(\frac{1}{x-1} - \frac{\frac{1}{2}(2x+2)-3}{x^2+2x+2} \right) dx$ $I = \int \left(\frac{1}{x-1} - \frac{1}{2} \times \frac{2x+2}{x^2+2x+2} + \frac{3}{(x+1)^2+1} \right) dx$ $I = \ln x-1 - \frac{1}{2} \ln(x^2+2x+2) + 3 \tan^{-1}(x+1) + c$ or $I = \ln \left \frac{x-1}{\sqrt{x^2+2x+2}} \right + 3 \tan^{-1}(x+1) + c$	3 marks – correct solution 2 marks – - writes the integrand as the correct sum of fractions and completes the square in a denominator, or equivalent merit 1 mark - correct integration achieving $\ln x-1$ - attempts to write as sum of three fractions.
b-i	$\begin{aligned} \omega^{2n} &= \left(e^{\pi i/n} \right)^{2n} \\ &= e^{2\pi i} \\ &= 1 \end{aligned}$	1 mark - Provides correct solution
b-ii	$\begin{aligned} \omega^{2n} - 1 &= 0 \\ \Rightarrow (\omega - 1)(\omega^{2n-1} + \dots + \omega + 1) &= 0 \\ \text{Since } \omega \neq 1, \omega^{2n-1} + \dots + \omega + 1 &= 0 \end{aligned}$	1 mark - Provides correct solution including statement that $\omega \neq 1$

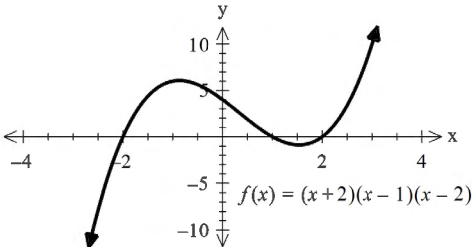
b-iii	$1 + \omega + \dots + \omega^{n-1} = \frac{1 - \omega^n}{1 - \omega}$ $= \frac{1 - e^{\pi i}}{1 - \cos\left(\frac{\pi}{n}\right) - i \sin\left(\frac{\pi}{n}\right)}$ Line 1 $= \frac{1 - (-1)}{1 - \cos\left(\frac{\pi}{n}\right) - i \sin\left(\frac{\pi}{n}\right)} \times \frac{1 - \cos\left(\frac{\pi}{n}\right) + i \sin\left(\frac{\pi}{n}\right)}{1 - \cos\left(\frac{\pi}{n}\right) + i \sin\left(\frac{\pi}{n}\right)}$ $= \frac{2 \left(1 - \cos\left(\frac{\pi}{n}\right) + i \sin\left(\frac{\pi}{n}\right)\right)}{\left(1 - \cos\left(\frac{\pi}{n}\right)\right)^2 + \left(\sin\left(\frac{\pi}{n}\right)\right)^2}$ Line 2 $= \frac{2 \left(1 - \cos\left(\frac{\pi}{n}\right) + i \sin\left(\frac{\pi}{n}\right)\right)}{2 - 2 \cos\left(\frac{\pi}{n}\right)}$ $= \frac{1 - \cos\left(\frac{\pi}{n}\right)}{1 - \cos\left(\frac{\pi}{n}\right)} + i \frac{\sin\left(\frac{\pi}{n}\right)}{1 - \cos\left(\frac{\pi}{n}\right)}$ $= 1 + i \frac{2 \sin\left(\frac{\pi}{2n}\right) \cos\left(\frac{\pi}{2n}\right)}{2 \sin^2\left(\frac{\pi}{2n}\right)}$ $= 1 + i \cot\left(\frac{\pi}{2n}\right)$	3 marks - Provides correct solution 2 marks - correct to Line 2 1 mark – Produces either fraction in Line 1.
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c	Suppose for a contradiction that a, b and c are odd integers and the equation $ax^2 + bx + c = 0$ has an integer solution. For $A, B, C \in \mathbb{Z}$, let $a = 2A + 1$, $b = 2B + 1$, and $c = 2C + 1$ $\therefore (2A + 1)x^2 + (2B + 1)x + (2C + 1) = 0$ [*] Case 1: x is an even number, i.e. $x = 2k$, $k \in \mathbb{Z}$. Since x is a solution, substitute into [*]: $(2A + 1)(2k)^2 + (2B + 1)(2k) + (2C + 1) = 0$ Expand to give: $2[2k^2(2A + 1) + k(2B + 1) + C] + 1 = 0$ LHS is an odd number which cannot equal to zero, hence absurd. Case 2: x is an odd number, i.e. $x = 2k + 1$, $k \in \mathbb{Z}$. Since x is a solution, substitute into [*]: $(2A + 1)(2k + 1)^2 + (2B + 1)(2k + 1) + (2C + 1) = 0$ Expand to give: $(2A + 1)(4k^2 + 4k + 1) + 4Bk + 2B + 2k + 1 + 2C + 1 = 0$ $2A(4k^2 + 4k) + 4k^2 + 6k + 3 + 4Bk + 2B + 2C = 0$ $2[A(4k^2 + 4k) + 2k^2 + 3k + 1 + 2Bk + B + C] + 1 = 0$ LHS is an odd number which cannot equal to zero, hence absurd. $\therefore$ initial supposition is false $\therefore$ if a, b and c are odd integers, then $ax^2 + bx + c = 0$ has no integer solution	4 marks – correct solution. 3 marks – attempts both odd and even expressions but only one correct. 2 marks - attempts one case only and proven correctly. - attempts to use quadratic formula to show no integer solutions ie. more than simply deriving quad formula 1 mark – correctly converts a, b and c to an expression for odd numbers.
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Question 13

a-i	$ \begin{aligned} 1 + i \cot \theta &= 1 + i \frac{\cos \theta}{\sin \theta} \\ &= \frac{\sin \theta + i \cos \theta}{\sin \theta} \\ &= \frac{1}{\sin \theta} [\sin \theta + i \cos \theta] \\ &= \frac{1}{\sin \theta} \left[\cos \left(\frac{\pi}{2} - \theta \right) + i \sin \left(\frac{\pi}{2} - \theta \right) \right] \\ &= \operatorname{cosec} \theta \operatorname{cis} \left(\frac{\pi}{2} - \theta \right) \end{aligned} $ $\sqrt{1 + \cot^2 \theta}$ was also accepted. If the argument was in degrees this was also accepted.	2 marks – correct solution 1 mark – either - correct modulus - correct argument
a-ii	$ \begin{aligned} (1 + i \cot \theta)^{-3} &= \left(\operatorname{cosec} \theta \operatorname{cis} \left(\frac{\pi}{2} - \theta \right) \right)^{-3} \\ &= \operatorname{cosec}^{-3} \theta \operatorname{cis} \left[-3 \left(\frac{\pi}{2} - \theta \right) \right] \\ &= \sin^3 \theta \operatorname{cis} \left(3\theta - \frac{3\pi}{2} \right) \\ &= \sin^3 \theta \operatorname{cis} \left(3\theta + \frac{\pi}{2} \right) \end{aligned} $	1 mark
b	Let $t = \tan \frac{x}{2}$ $dt = \frac{1}{2} \sec^2 \frac{x}{2} dx = \frac{1}{2} (1 + t^2) dx$ $dx = \frac{2}{1+t^2} dt$ $\therefore \sin x = \frac{2t}{1+t^2}$ When $x = 0, t = 0$ $x = \frac{\pi}{2}, t = 1$ $\therefore I = \int_0^{\frac{\pi}{2}} \frac{1}{2 - \sin x} dx = \int_0^1 \frac{1}{2 - \frac{2t}{1+t^2}} \frac{2}{1+t^2} dt$ $I = \int_0^1 \frac{2}{2 - \frac{2t}{1+t^2}} \times \frac{2}{1+t^2} dt$ $I = \int_0^1 \frac{2}{2(1+t^2) - 2t} dt$ $I = \int_0^1 \frac{2}{2t^2 - 2t + 2} dt$	4 marks - Provides correct solution 3 marks - Finds the correct integral, or equivalent merit 2 marks - Completes t-substitution, including the limits of integration, or equivalent merit 1 mark - Attempts to use a t-substitution, or equivalent merit

	$= \int_0^1 \frac{1}{t^2 - t + 1} dt$ $= \int_0^1 \frac{1}{\left(t - \frac{1}{2}\right)^2 + \frac{3}{4}} dt$ $= \left[\frac{2}{\sqrt{3}} \tan^{-1} \left(\frac{2t - 1}{\sqrt{3}} \right) \right]_0^1$ $= \frac{2}{\sqrt{3}} \tan^{-1} \frac{1}{\sqrt{3}} - \frac{2}{\sqrt{3}} \tan^{-1} \left(-\frac{1}{\sqrt{3}} \right)$ $= \frac{2}{\sqrt{3}} \times \frac{\pi}{6} - \frac{2}{\sqrt{3}} \times -\frac{\pi}{6}$ $= \frac{4\pi}{6\sqrt{3}}$ $= \frac{2\pi}{3\sqrt{3}}$ $= \frac{2\sqrt{3}\pi}{9}$	
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c	$P(n): u_n = 3^n - n - 1, n \in \mathbb{Z}^+,$ for the sequence $u_1 = 1, u_{n+1} = 3u_n + 2n + 1, n \in \mathbb{Z}^+.$ Prove $P(1)$ true: By formula, $u_1 = 3^1 - 1 - 1 = 1$ By definition, $u_1 = 1$ $\therefore P(1)$ is true Assume $P(k)$ true, $k \in \mathbb{Z}^+$ i.e. $u_k = 3^k - k - 1$ Prove $P(k+1)$ true. i.e. prove $u_{k+1} = 3^{k+1} - (k+1) - 1 = 3^{k+1} - k - 2$ Now, from the definition: $u_{k+1} = 3u_k + 2k + 1$ $u_{k+1} = 3(3^k - k - 1) + 2k + 1$ using the assumption $u_{k+1} = 3^{k+1} - 3k - 3 + 2k + 1$ $u_{k+1} = 3^{k+1} - k - 2$ $\therefore P(k+1)$ is true if $P(k)$ is true. Hence, by mathematical induction, $P(n)$ is true for all positive integers $n \geq 1$	3 marks – correct solution 2 marks - Proves true for $n = 1$ and incorporates the assumption $P(k)$ into $P(k+1)$ 1 mark - Proves true for $n = 1$
d-i	$\frac{d}{dx} \left(\frac{1}{2} v^2 \right) = 4(x-2)^{-2} + 8x^{-3}$ $\frac{1}{2} v^2 = -4(x-2)^{-1} - 4x^{-2} + c$ $x=1, v=0 \Rightarrow c=0$ $\therefore \frac{1}{2} v^2 = -4(x-2)^{-1} - 4x^{-2}$ $\therefore v^2 = -\frac{8}{x-2} - \frac{8}{x^2}$	2 marks - Provides correct solution 1 mark - Makes some use of $\frac{d}{dx} \left(\frac{1}{2} v^2 \right)$ or equivalent merit
d-ii	 $v^2 \geq 0$ $\Rightarrow -8 \left(\frac{1}{x-2} + \frac{1}{x^2} \right) \geq 0$ $\Rightarrow \frac{1}{x-2} + \frac{1}{x^2} \leq 0$ $\Rightarrow \frac{x^2 + x - 2}{x^2(x-2)} \leq 0$ $\Rightarrow \frac{(x+2)(x-1)(x-2)}{x^2(x-2)^2} \leq 0$ From the sketch of the cubic numerator, $x \leq -2$ or $1 \leq x < 2$ [Note that $x-2$ is originally in the denominator] Since the particle moves continuously, we can only use $1 \leq x < 2$	3 marks – correct solution 2 marks – determine both possible solutions but does not exclude $x=2$ 1 mark – attempts to solve required inequality.

OR

$$\frac{x^2 + x - 2}{x^2(x - 2)} = 0 \quad x \neq 0 \text{ or } x \neq 2$$

$$x^2 + x - 2 = 0$$

$$(x + 2)(x - 1) = 0$$

$$x = -2 \text{ or } x = 1$$

Test points: when $x < -2$ inequality is true

when $-2 < x < 0$ inequality is false

when $0 < x < 1$ inequality is false

when $1 < x < 2$ inequality is true

when $x > 2$ inequality is false

since the particle starts at $x = 1$ and moves continuously
only solution is $1 \leq x < 2$

However, students who left $x \leq -2$ were not penalised.

Question 14

a	$I = \int_0^4 \frac{\sqrt{x}}{(x\sqrt{x} + 1)^2} dx$ Let $u^2 = x$ ($u \geq 0$) è $2u du = dx$ When $x = 0, u = 0$ $x = 4, u = 2$ $\therefore I = \int_0^2 \frac{u}{(u^2 \cdot u + 1)^2} 2u du = \int_0^2 \frac{2u^2}{(u^3 + 1)^2} du$ $I = \left[-\frac{2}{3(u^3 + 1)} \right]_0^2 = -\frac{2}{3 \times 9} - \left(-\frac{2}{3} \right) = \frac{16}{27}$	3 marks - correct solution 2 marks - Obtains a correct primitive, or equivalent merit 1 mark - Uses an appropriate substitution, including the limits of integration, or equivalent merit
b-i	If x, y are two integers for which one is odd and one is even, then $x + y$ is odd.	1 mark – correct solution
b-ii	Assume x is even, y is odd $\exists k, m \in \mathbb{Z} \text{ such } x = 2k, y = 2m + 1$ $x + y = 2k + 2m + 1$ $= 2(k + m) + 1$ $\therefore x + y$ is odd $nQ \Rightarrow nP$	2 marks – correct solution 1 mark – attempts correct method ie. makes correct statement $\exists k, m \in \mathbb{Z} \text{ such } x = 2k, y = 2m + 1$
c-i	$\frac{1}{a} + \frac{1}{4-a} - 1 = \frac{4-a+a-a(4-a)}{a(4-a)}$ $= \frac{(a-2)^2}{a(4-a)} \geq 0 \text{ since } 0 < a < 4 \text{ makes the denominator positive}$ Hence $\frac{1}{a} + \frac{1}{4-a} \geq 1$	2 marks – correct solution 1 mark – forms correct inequality
c-ii	Assume $\frac{1}{a} + \frac{1}{4-b} < 1, \frac{1}{b} + \frac{1}{4-c} < 1$ and $\frac{1}{c} + \frac{1}{4-a} < 1$ By addition, $\left(\frac{1}{a} + \frac{1}{4-b} \right) + \left(\frac{1}{b} + \frac{1}{4-c} \right) + \left(\frac{1}{c} + \frac{1}{4-a} \right) < 3$ $\Rightarrow \left(\frac{1}{a} + \frac{1}{4-a} \right) + \left(\frac{1}{b} + \frac{1}{4-b} \right) + \left(\frac{1}{c} + \frac{1}{4-c} \right) < 3$ But, from part (i), each bracket is at least 1, a contradiction. Hence, at least one of the three numbers is at least 1	3 marks – correct solution 2 marks - Makes significant progress towards a solution. 1 mark - Makes little progress towards a solution

	Option 1 $I_n = \int_0^1 \frac{x^{2n}}{\sqrt{1+x^2}} dx$ $I_n = \int_0^1 \frac{x^2 x^{2n-2}}{\sqrt{1+x^2}} dx \text{ since } n \geq 2$ $I_n = \int_0^1 \frac{(1+x^2-1)x^{2n-2}}{\sqrt{1+x^2}} dx$ $I_n = \int_0^1 \frac{(1+x^2)x^{2n-2}}{\sqrt{1+x^2}} dx - \int_0^1 \frac{x^{2(n-1)}}{\sqrt{1+x^2}} dx$ $I_n = \int_0^1 x^{2n-2} \sqrt{1+x^2} dx - I_{n-1}$ d For $\int_0^1 x^{2n-2} \sqrt{1+x^2} dx$, integrate by parts as follows: $u = \sqrt{1+x^2} \quad \therefore \frac{du}{dx} = \frac{x}{\sqrt{1+x^2}}$ $\frac{dv}{dx} = x^{2n-2} \quad \therefore v = \frac{1}{2n-1} x^{2n-1}$ $\therefore I_n = \left[\frac{x^{2n-1} \sqrt{1+x^2}}{2n-1} \right]_0^1 - \frac{1}{2n-1} \int_0^1 x^{2n-1} \frac{x}{\sqrt{1+x^2}} dx - I_{n-1}$ $I_n = \frac{\sqrt{2}}{2n-1} - \frac{1}{2n-1} \int_0^1 \frac{x^{2n}}{\sqrt{1+x^2}} dx - I_{n-1}$ $I_n = \frac{\sqrt{2}}{2n-1} - \frac{1}{2n-1} I_n - I_{n-1}$ $(2n-1)I_n = \sqrt{2} - I_n - (2n-1)I_{n-1}$ $\therefore 2n I_n + (2n-1)I_{n-1} = \sqrt{2} \text{ as required}$	4 marks – correct solution. 3 marks - Makes substantial progress, identifying the terms I_n and I_{n-1} 2 marks - Attempts to use integration by parts and attempts a manipulation to obtain I_{n-1} or I_n, or equivalent merit 1 mark - Attempts to use integration by parts, or equivalent merit
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Option 2

$$I_n = \int_0^1 \frac{x^{2n}}{\sqrt{1+x^2}} dx$$

$$= \int_0^1 \frac{x \cdot x^{2n-1}}{\sqrt{1+x^2}} dx$$

$$u = x^{2n-1} \quad dv = \frac{x}{\sqrt{1+x^2}}$$

$$du = (2n-1)x^{2n-2} \quad v = \int \frac{x}{\sqrt{1+x^2}} dx$$

$$= \frac{1}{2} \int \frac{2x}{\sqrt{1+x^2}} dx$$

$$= (1+x^2)^{\frac{1}{2}} = \sqrt{1+x^2}$$

$$I_n = \left[x^{2n-1} \sqrt{1+x^2} \right]_0^1 - \int_0^1 \sqrt{1+x^2} \cdot (2n-1) \cdot x^{2n-2} dx$$

d.

$$= \sqrt{2} - \int_0^1 \sqrt{1+x^2} \cdot (2n-1) \cdot x^{2n-2} dx$$

$$= \sqrt{2} - (2n-1) \cdot \int_0^1 \sqrt{1+x^2} \cdot x^{2n-2} dx$$

$$= \sqrt{2} - (2n-1) \cdot \int_0^1 \frac{(1+x^2) \cdot x^{2n-2}}{\sqrt{1+x^2}} dx$$

$$= \sqrt{2} - (2n-1) \left[\int_0^1 \frac{x^{2n-2}}{\sqrt{1+x^2}} dx + \int_0^1 \frac{x^2}{\sqrt{1+x^2}} dx \right]$$

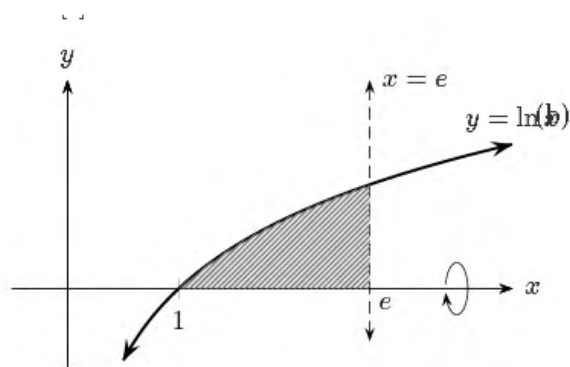
$$= \sqrt{2} - (2n-1)[I_{n-1} + I_n]$$

$$= \sqrt{2} - (2n-1)I_{n-1} - (2n-1)I_n]$$

$$I_n + 2nI_n - I_n + (2n-1)I_{n-1} = \sqrt{2}$$

$$2nI_n + (2n-1)I_{n-1} = \sqrt{2}$$

Question 15



$$V = \pi \int_1^e y^2 dx$$

$$= \pi \int_1^e (\ln x)^2 dx$$

Finding the integral by parts/inserting
'phantom' term

a

$$\left| \begin{array}{ll} u = (\ln x)^2 & dv = 1 \\ du = 2 \ln x \times \frac{1}{x} & v = x \\ & = \frac{2}{x} \ln x \end{array} \right.$$

$$V = \pi \left(\left[x (\ln x)^2 \right]_1^e - \int_1^e \frac{2}{x} \ln x \times x dx \right)$$

$$= \pi \left(e - 2 \int_1^e \ln x dx \right)$$

Use the area about the y axis:

$$\left| \begin{array}{l} \int_1^e \ln x dx = (e \times 1) - \int_{y=0}^{y=1} e^y dy \\ = e - [e - 1] \\ = 1 \\ \therefore V = \pi(e - 2) \text{ units}^3 \end{array} \right.$$

3 marks – correct solution

2 marks – correct use of
IBP

1 mark

b-i	Let $\underline{r} = x\underline{i} + y\underline{j} + z\underline{k}$, $\left \underline{r} - (3\underline{i} + \underline{j} + 4\underline{k}) \right = \sqrt{35}$ $\left (x-3)\underline{i} + (y-1)\underline{j} + z-4\underline{k} \right = \sqrt{35}$ $\therefore (x-3)^2 + (y-1)^2 + (z-4)^2 = 35$	1 mark
b-i	$\underline{r} = \begin{pmatrix} 2 \\ 0 \\ 3 \end{pmatrix} + \lambda \begin{pmatrix} 1 \\ 2 \\ -1 \end{pmatrix}$ If the line is a tangent, then only one unique value of λ exists for the expression $\left \underline{r} - \begin{pmatrix} 3 \\ 1 \\ 4 \end{pmatrix} \right = \sqrt{35}$ Test for values of λ that satisfy following $\left \underline{r} - \begin{pmatrix} 3 \\ 1 \\ 4 \end{pmatrix} \right = \sqrt{35}$ $\text{LHS} = \left \begin{pmatrix} 2 \\ 0 \\ 3 \end{pmatrix} + \lambda \begin{pmatrix} 1 \\ 2 \\ -1 \end{pmatrix} - \begin{pmatrix} 3 \\ 1 \\ 4 \end{pmatrix} \right $ $= \left \begin{pmatrix} -1 + \lambda \\ -1 + 2\lambda \\ -1 - \lambda \end{pmatrix} \right $ $= \sqrt{(\lambda - 1)^2 + (2\lambda - 1)^2 + (-1 - \lambda)^2}$ $= \sqrt{\lambda^2 - 2\lambda + 1 + 4\lambda^2 - 4\lambda + 1 + \lambda^2 + 2\lambda + 1}$ $= \sqrt{6\lambda^2 - 4\lambda + 3} = \sqrt{35}$ $6\lambda^2 - 4\lambda + 3 = 35$ $6\lambda^2 - 4\lambda - 32 = 0$ $3\lambda^2 - 2\lambda - 16 = 0$ $\Delta = (-2)^2 - 4(-16)(3)$ $= 196 > 0$ $\therefore$ 2 solutions $\Rightarrow$ not a tangent	3 marks

Show z, w exist given $\frac{1}{z} + \frac{1}{w} = \frac{1}{z+w}$

$\therefore$

$$\frac{1}{z} + \frac{1}{w} = \frac{1}{z+w}$$

$$\frac{z+w}{z} + \frac{z+w}{w} = 1$$

$$\frac{z}{z} + \frac{w}{z} + \frac{z}{w} + \frac{w}{w} = 1$$

$$1 + \frac{z}{w} + \frac{w}{z} + 1 = 1$$

$$\frac{z}{w} + \frac{w}{z} + 1 = 0 \quad \therefore \text{Line 1}$$

Let $k = \left(\frac{z}{w}\right)$

Line 1 becomes

$$k + \frac{1}{k} + 1 = 0$$

$$k^2 + 1 + k = 0$$

$$k^2 + k + 1 = 0$$

$$k^2 + k = -1$$

$$k^2 + k + \frac{1}{4} + \frac{3}{4} = 0$$

$$\left(k + \frac{1}{2}\right)^2 + \frac{3}{4} = 0$$

$$\left(k + \frac{1}{2}\right)^2 = -\frac{3}{4}$$

$$k = -\frac{1}{2} \pm \frac{\sqrt{3}}{2}i$$

$\therefore$ as k exists as a complex number

then $\frac{z}{w}$ exists as complex number

hence complex numbers z and w exist

such that $\frac{1}{z} + \frac{1}{w} = \frac{1}{z+w}$

eg $z = -1 + 3i$ and $w = 2$

3 marks – correct solution

2 marks – simplifies the question by assigning a value for either z or w

1 mark – correct solution for assumption that $|z| = |w| = 1$

d-i	$a^2 + b^2 + c^2 \geq ab + ac + bc$ $(a - b)^2 \geq 0$ $a^2 - 2ab + b^2 \geq 0$ $a^2 + b^2 \geq 2ab$ Similarly $a^2 + c^2 \geq 2ac$ $b^2 + c^2 \geq 2bc$ $2a^2 + 2b^2 + 2c^2 \geq 2ab + 2ac + 2bc$ $a^2 + b^2 + c^2 \geq ab + ac + bc$	1 mark
e-i	$(a + b + c)^2 \geq 3(ab + ac + bc)$ $\text{LHS} = (a + b + c)^2$ $= a^2 + b^2 + c^2 + 2ab + 2bc + 2ac$ $= a^2 + b^2 + c^2 + 2ab + 2bc + 2ac$ $\geq ab + bc + ac + 2ab + 2bc + 2ac$ $\geq 3ab + 3bc + 3ac$ $\geq 3(ab + bc + ac)$	1 mark

e-ii	<i>hence prove: $a^2 + b^2 + c^2 \geq 3\sqrt[3]{a^2b^2c^2}$</i> $(a - b)^2 \geq 0$ $a^2 - 2ab + b^2 \geq 0$ $a^2 - ab + b^2 \geq ab$ $a^3 + b^3 = (a + b)(a^2 - ab + b^2)$ $a^3 + b^3 \geq (a + b)ab \dots (1)$ <i>Similarly</i> $b^3 + c^3 \geq (c + b)cb \dots (2)$ $a^3 + c^3 \geq (a + c)ac \dots (3)$ <i>Multiply Lines 1, 2, 3 by $\frac{c}{c}, \frac{a}{a}$ and $\frac{b}{b}$ gives</i> $a^3 + b^3 \geq \left(\frac{a}{c} + \frac{b}{c}\right)abc$ $b^3 + c^3 \geq \left(\frac{b}{a} + \frac{c}{a}\right)abc$ $a^3 + c^3 \geq \left(\frac{a}{b} + \frac{c}{b}\right)abc$ <i>Now add these new inequalities</i> $2(a^3 + b^3 + c^3) \geq abc\left(\frac{a}{c} + \frac{b}{c} + \frac{b}{a} + \frac{c}{a} + \frac{a}{b} + \frac{c}{b}\right)$ $2(a^3 + b^3 + c^3) \geq abc\left(\frac{a}{b} + \frac{b}{a} + \frac{a}{c} + \frac{c}{a} + \frac{b}{c} + \frac{c}{b}\right)$ $2(a^3 + b^3 + c^3) \geq abc\left(\frac{a^2 + b^2}{ab} + \frac{a^2 + c^2}{ac} + \frac{b^2 + c^2}{bc}\right)$ $nb \ a^2 + b^2 \geq 2ab$ $\therefore 2(a^3 + b^3 + c) \geq abc\left(\frac{2ab}{ab} + \frac{2ac}{ac} + \frac{2bc}{bc}\right)$ $2(a^3 + b^3 + c^3) \geq 6abc$ $a^3 + b^3 + c^3 \geq 3abc$ $\text{Let } a \rightarrow \sqrt[3]{a} \quad b \rightarrow \sqrt[3]{b} \quad c \rightarrow \sqrt[3]{c}$ $a + b + c \geq 3\sqrt[3]{abc}$ $\text{Let } a \rightarrow \sqrt[3]{a^2} \quad b \rightarrow \sqrt[3]{b^2} \quad c \rightarrow \sqrt[3]{c^2}$ $a^2 + b^2 + c^2 \geq 3\sqrt[3]{a^2b^2c^2}$	3 marks - correct 2 marks – correct however used AM v GM (Note: if “Hence” had not been used in the question – it would have been legitimate to use AMvGM statement. 1 mark – some progress using part i and ii
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Question 16

a-i	Clearly, $z = -1 = \text{cis } \pi$ is a solution. The solutions lie on the unit circle, equally spaced. Hence, solutions are: $z = \text{cis } \pi, \text{cis } \frac{\pi}{3}, \text{cis } \left(-\frac{\pi}{3}\right)$	2 marks – correct solution 1 mark - Writes one of the solutions in modulus-argument form
a-ii	Using part (ii), $z^2 + 2 = \text{cis } \pi, \text{cis } \frac{\pi}{3}, \text{cis } \left(-\frac{\pi}{3}\right)$ $z^2 + 2 = \text{cis } \pi$: $z^2 + 2 = -1$ $z^2 = -3$ $z = \pm i\sqrt{3}$ $z^2 + 2 = \text{cis } \frac{\pi}{3}$: $z^2 + 2 = \frac{1}{2} + \frac{\sqrt{3}}{2}i$ $z^2 = -\frac{3}{2} + \frac{\sqrt{3}}{2}i$ $z^2 = \sqrt{3} \text{cis } \frac{5\pi}{6}$ $z = \sqrt[4]{3} \text{cis } \frac{5\pi}{12}, \sqrt[4]{3} \text{cis } \left(\frac{5\pi}{12} - \pi\right)$ $z = \sqrt[4]{3} \text{cis } \frac{5\pi}{12}, \sqrt[4]{3} \text{cis } \left(-\frac{7\pi}{12}\right)$ $z^2 + 2 = \text{cis } \left(-\frac{\pi}{3}\right)$: $z^2 + 2 = \frac{1}{2} - \frac{\sqrt{3}}{2}i$ $z^2 = -\frac{3}{2} - \frac{\sqrt{3}}{2}i$ $z^2 = \sqrt{3} \text{cis } \left(-\frac{5\pi}{6}\right)$ $z = \sqrt[4]{3} \text{cis } \left(-\frac{5\pi}{12}\right), \sqrt[4]{3} \text{cis } \left(-\frac{5\pi}{12} + \pi\right)$ $z = \sqrt[4]{3} \text{cis } \left(-\frac{5\pi}{12}\right), \sqrt[4]{3} \text{cis } \left(\frac{7\pi}{12}\right)$ $\therefore$ The six solutions are: $z = \pm i\sqrt{3}, \sqrt[4]{3} \text{cis } \frac{5\pi}{12}, \sqrt[4]{3} \text{cis } \left(-\frac{5\pi}{12}\right), \sqrt[4]{3} \text{cis } \frac{7\pi}{12}, \sqrt[4]{3} \text{cis } \left(-\frac{7\pi}{12}\right)$	3 marks – correct solution 2 marks - Finds two solutions, each from a different substitution of the 3 expressions from (i), or equivalent merit. Equivalent forms of solutions were accepted. 1 mark - Finds one solution, or equivalent merit

b-i	Find $\overrightarrow{PR}$ and $\overrightarrow{QS}$ in terms of e. $\overrightarrow{PR} = \overrightarrow{OR} - \overrightarrow{OP}$ $\overrightarrow{OR} = \frac{\overrightarrow{OE} + \overrightarrow{OC}}{2} = \frac{4e\mathbf{k} + 4\mathbf{i} + 4\mathbf{j}}{2} = 2\mathbf{i} + 2\mathbf{j} + 2e\mathbf{k}$ $\overrightarrow{OP} = \frac{\overrightarrow{OA} + \overrightarrow{OB}}{2} = -4\mathbf{j}$ $\therefore \overrightarrow{PR} = 2\mathbf{i} + 6\mathbf{j} + 2e\mathbf{k}$ Similarly, $\overrightarrow{QS} = -6\mathbf{i} - 2\mathbf{j} + 2e\mathbf{k}$	2 Marks: Provides correct solution. (Must include working for QS) 1 Mark: Finds $\overrightarrow{PR}$ or $\overrightarrow{QS}$, or equivalent merit
b-ii	For Line l, use point P and direction vector $\overrightarrow{PR}$: $\therefore \text{Line } l: \begin{pmatrix} x \\ y \\ z \end{pmatrix} = \begin{pmatrix} 0 \\ -4 \\ 0 \end{pmatrix} + \lambda \begin{pmatrix} 2 \\ 6 \\ 2e \end{pmatrix}$ For Line s, use point Q and direction vector $\overrightarrow{QS}$: $\therefore \text{Line } s: \begin{pmatrix} x \\ y \\ z \end{pmatrix} = \begin{pmatrix} 4 \\ 0 \\ 0 \end{pmatrix} + \mu \begin{pmatrix} -6 \\ -2 \\ 2e \end{pmatrix}$	2 Marks: Provides correct solution. OR Obtains equations consistent with an error in part (i) 1Mark: Writes the vector equation of either Line l or Line s, or equivalent merit OR Obtains the correct direction vector for each line.
b-iii	To find the point of intersection, equate the x components: $2\lambda = 4 - 6\mu$ $\therefore \lambda = 2 - 3\mu \quad [1]$ Equate the y components and use [1]: $-4 + 6\lambda = -2\mu$ $-4 + 6(2 - 3\mu) = -2\mu$ $\therefore \mu = \frac{1}{2} \quad \text{and} \quad \lambda = \frac{1}{2}$ $\therefore$ Point of intersection is $(1, -1, e)$ $\therefore \overrightarrow{OM} = \mathbf{i} - \mathbf{j} + e\mathbf{k}$	2 Marks - Uses the values of the parameters to find the point of intersection, making one error 1 Mark - Correctly equates x and y components and attempts to solve simultaneously to find the values of the parameters
b-iv	$\overrightarrow{EB} = \overrightarrow{OB} - \overrightarrow{OE} = 4\mathbf{i} - 4\mathbf{j} - 4e\mathbf{k}$ If $\overrightarrow{OM} \perp \overrightarrow{EB}$, $\overrightarrow{OM} \cdot \overrightarrow{EB} = 0$:	3 Marks: Correct answer 2 Marks: Obtains two correct values for e, $\cos\theta$ or θ

$\begin{pmatrix} 1 \\ -1 \\ e \end{pmatrix} \cdot \begin{pmatrix} 4 \\ -4 \\ -4e \end{pmatrix} = 0$ $4 + 4 - 4e^2 = 0$ $e = \sqrt{2} \quad (e > 0)$ The angle between $\overrightarrow{PR}$ and $\overrightarrow{QS}$: $\overrightarrow{PR} = 2\mathbf{i} + 6\mathbf{j} + 2\sqrt{2}\mathbf{k}$ $\overrightarrow{QS} = -6\mathbf{i} - 2\mathbf{j} + 2\sqrt{2}\mathbf{k}$ $\therefore \cos \theta = \frac{\overrightarrow{PR} \cdot \overrightarrow{QS}}{ \overrightarrow{PR} \overrightarrow{QS} } = \frac{-12 - 12 + 8}{\left(\sqrt{2^2 + 6^2 + (2\sqrt{2})^2}\right)^2} = -\frac{1}{3}$ $\therefore \theta \cong 109^\circ \text{ (nearest degree)}$	OR Obtains values of $\cos \theta$ and θ consistent with an incorrect value of e. 1 Mark: Finds a correct expression for vector EB OR Finds the value of e OR Finds an angle consistent with incorrect values for EB and e.
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